

GitHub Agentic Efficiency Improver: Quickstart Guide

Set up automated energy efficiency improvements for your codebase in under 10 minutes using GitHub Agentic Workflows.

Prerequisites

Before you start, verify you have:

- **GitHub CLI v2.0.0+** — Check with `gh --version`
 - **GitHub account** with write access to a repository
 - **GitHub Actions enabled** in your repository
 - **AI provider account** — Choose one:
 - GitHub Copilot (recommended)
 - Claude (Anthropic)
 - OpenAI
 - Google Gemini
 - **API key or personal access token** ready for your chosen AI provider
 - **Test repository** with some commit history (for meaningful optimization results)
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Setup & Run in 5 Steps

1. Install the GitHub Agentic Workflows Extension

```
gh extension install github/gh-aw
```

Verify installation:

```
gh aw --help
```

2. Navigate to Your Repository & Launch the Wizard

```
cd your-repository
gh aw add-wizard githubnext/agentics/efficiency-improver
```

The wizard will guide you through configuration. No manual file editing needed.

3. Select Your AI Provider & Add Credentials

When prompted, choose your AI provider:

- **GitHub Copilot** (integrates natively with GitHub)
- **Claude** (Anthropic)
- **OpenAI**
- **Google Gemini**

Then provide your API credentials. The wizard automatically stores them as repository secrets—they're never exposed in logs or workflow files.

4. Review & Merge the Pull Request

The wizard creates a pull request with:

- `.github/workflows/efficiency-improver.lock.yml` — The compiled GitHub Actions workflow
- `efficiency-improver.md` — The markdown source file

Review the PR, then **merge it**. The workflow will automatically start running.

5. Monitor & Review Results

1. Go to the **Actions** tab in your repository
2. Watch the `Efficiency Improver` workflow run (takes up to 60 minutes)
3. Once complete, check the **Pull Requests** tab for optimization suggestions
4. Review generated PRs with before/after metrics on energy efficiency

What Gets Generated

- **Pull Requests** with concrete code optimizations and energy savings metrics
- **Comments on Issues** with AI-driven efficiency suggestions
- **Activity Summaries** tracking optimization progress and energy savings
- **Transparent Attribution** clearly disclosing AI involvement and methodology

Workflow Configuration (Optional)

The generated `efficiency-improver.md` file contains:

Frontmatter (YAML):

```
on:
  schedule:
    - cron: '0 9 * * *'           # Runs daily at 9 AM (edit to customize)
  workflow_dispatch: {}         # Trigger manually when needed
agents: [github-copilot]        # Change if using different AI provider
read_permissions:
  - contents: read
  - issues: read
  - pull_requests: read
safe_outputs:
  - pull_requests: write
  - issues: write
  - contents: write
```

To customize:

1. Edit the markdown file
2. If you change frontmatter, run: `gh aw compile` to regenerate the lock file
3. Commit and push your changes

Common customizations:

- **Change schedule:** Edit the cron expression (use crontab.guru for help)

- **Switch AI provider:** Change the `agents` field and update your API credentials in repository secrets
- **Modify permissions:** Edit `read_permissions` or `safe_outputs` (use minimal necessary permissions)

Troubleshooting

Issue	Solution
gh aw command not found	Update GitHub CLI: <code>gh extension upgrade gh-aw</code> or reinstall with <code>gh extension install github/gh-aw</code>
Wizard doesn't start	Run <code>gh auth status</code> to verify authentication; re-authenticate if needed with <code>gh auth login</code>
Workflow fails in Actions tab	Check repository secrets: Go to Settings > Secrets and variables > Actions . Verify API key is named correctly (e.g., <code>ANTHROPIC_API_KEY</code> , <code>OPENAI_API_KEY</code>). Click the failed workflow and then click the section that failed and it most likely will tell you what the issue is below; also you can also ask GitHub Copilot what this issue is by clicking the 'Explain error' button.
Workflow takes too long	Efficiency Improver can take up to 60 minutes.
No PRs generated after run	Check workflow logs in Actions tab. May need multiple runs to find optimization opportunities. Repos with complex patterns yield better results
Want to re-run the workflow	Delete generated PRs/issues and manually trigger: <code>gh workflow run efficiency-improver.yml</code>
Permission denied when pushing	Verify your GitHub token has repo and workflow scopes: <code>gh auth status</code>

Need Help?

- [GitHub Agentic Workflows Documentation](#) — Full documentation and examples
- [Efficiency Improver Workflow](#) — Learn how the workflow scans your code
- **GitHub CLI Docs** — `gh aw --help` or visit cli.github.com